

5G RAN

Adaptive Managed Latency Feature Parameter Description

Issue	Draft A
Date	2025-12-31



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1

Change History

1

2

About This Document

2

2.1

General Statements

2

2.2

Features in This Document

2

2.3

Differences

3

3

Overview

5

4

L4S Congestion Control

6

4.1

L4S Congestion Control Based on Local Identification

6

4.1.1

Principles

6

4.1.2

Network Analysis

8

4.1.2.1

Benefits

9

4.1.2.2

Impacts

9

4.1.3

Requirements

9

4.1.3.1

Licenses

9

4.1.3.2

Functions

9

4.1.3.2.1

Prerequisite Functions

10

4.1.3.2.2

Mutually Exclusive Functions

10

4.1.3.2.3

Function Impacts

10

4.1.3.3

Hardware

10

4.1.3.4

Others

10

4.1.4

Operation and Maintenance

11

4.1.4.1

Data Configuration

11

4.1.4.1.1

Data Preparation

11

4.1.4.1.2

Using MML Commands

12

4.1.4.1.3

Using the MAE-Deployment

13

4.1.4.2

Activation Verification

13

4.1.4.3

Network Monitoring

14

4.2

Enhanced L4S Congestion Control

14

4.2.1

Principles

14

4.2.2

Network Analysis

18

4.2.2.1

Benefits

18

4.2.2.2

Impacts

18

4.2.3 Requirements.....	18
4.2.3.1 Licenses.....	18
4.2.3.2 Functions.....	19
4.2.3.2.1 Prerequisite Functions.....	19
4.2.3.2.2 Mutually Exclusive Functions.....	19
4.2.3.2.3 Function Impacts.....	19
4.2.3.3 Hardware.....	19
4.2.3.4 Others.....	19
4.2.4 Operation and Maintenance.....	19
4.2.4.1 Data Configuration.....	19
4.2.4.1.1 Data Preparation.....	19
4.2.4.1.2 Using MML Commands.....	22
4.2.4.1.3 Using the MAE-Deployment.....	23
4.2.4.2 Activation Verification.....	23
4.2.4.3 Network Monitoring.....	24
4.3 Enhanced L4S Congestion Control Based on Rate Adjustment.....	24
4.3.1 Principles.....	24
4.3.2 Network Analysis.....	29
4.3.2.1 Benefits.....	29
4.3.2.2 Impacts.....	30
4.3.3 Requirements.....	30
4.3.3.1 Licenses.....	30
4.3.3.2 Functions.....	30
4.3.3.2.1 Prerequisite Functions.....	30
4.3.3.2.2 Mutually Exclusive Functions.....	31
4.3.3.2.3 Function Impacts.....	31
4.3.3.3 Hardware.....	33
4.3.3.4 Others.....	33
4.3.4 Operation and Maintenance.....	33
4.3.4.1 Data Configuration.....	33
4.3.4.1.1 Data Preparation.....	33
4.3.4.1.2 Using MML Commands.....	35
4.3.4.1.3 Using the MAE-Deployment.....	37
4.3.4.2 Activation Verification.....	38
4.3.4.3 Network Monitoring.....	38
5 Parameters.....	39
6 Counters.....	41
7 Glossary.....	42
8 Reference Documents.....	43

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue is the first release.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-071203	Adaptive Managed Latency	4 L4S Congestion Control

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
L4S congestion control based on local identification	Supported in both NSA and SA networking, with the following differences: • The function described in this document is supported only in SA networking. For details about this function in NSA networking, see <i>EPC-based NSA Basics</i>. • For NSA UEs, uplink and downlink congestion identification are supported only in Option 3x networking. • In SA networking, the congestion check period can be set to a minimum of 1 ms.	4.1 L4S Congestion Control Based on Local Identification
Enhanced L4S congestion control	This function takes effect only for SA UEs.	4.2 Enhanced L4S Congestion Control
Enhanced L4S congestion control based on rate adjustment	This function takes effect only for SA UEs.	4.3 Enhanced L4S Congestion Control Based on Rate Adjustment

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
L4S congestion control based on local identification	Supported only in low frequency bands	4.1 L4S Congestion Control Based on Local Identification
Enhanced L4S congestion control	Supported only in low frequency bands	4.2 Enhanced L4S Congestion Control
Enhanced L4S congestion control based on rate adjustment	Supported only in low frequency bands	4.3 Enhanced L4S Congestion Control Based on Rate Adjustment

3 Overview

Rapid development has been seen in technologies such as real-time media, remote control, industrial control, and autonomous driving. Base stations must provide better latency assurances for services including multiplayer gaming, cloud gaming, tele-operated driving, automated guided vehicles (AGVs), drones, video conferencing, cloud VR, and cloud AR.

These services require high latency stability due to their high traffic consumption and rate adjustment ability. To prevent large fluctuation in service latency as a result of congestion, Huawei has introduced the Adaptive Managed Latency feature. This feature provides the following functions: L4S congestion control based on local identification, enhanced L4S congestion control, and enhanced L4S congestion control based on rate adjustment.

Before enabling the sub-functions of L4S congestion control, set the **gNBQciBearer.L4sFlag** parameter to **TRUE**.

4 L4S Congestion Control

4.1 L4S Congestion Control Based on Local Identification

4.1.1 Principles

L4S congestion control based on local identification is used to relieve congestion and improve user experience for services such as cloud gaming and extended reality (XR). In SA networking, this function works as follows:

1. Specific 5QIs are configured for these services on the core network.
2. The gNodeB identifies the uplink and downlink service latencies caused by congestion based on the specific 5QIs.

L4S congestion control based on local identification includes the **uplink L4S congestion control** and **downlink L4S congestion control** functions. Before enabling uplink L4S congestion control or downlink L4S congestion control, enable the L4S function by setting **gNBQciBearer.L4sFlag** to **TRUE**.

For details about L4S congestion control in NSA Option 3x networking, see *EPC-based NSA Basics*.

Uplink L4S Congestion Control

If uplink congestion control is required, enable the uplink L4S congestion control function, which is controlled by the **UL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter. This function also requires that the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter be set to **L4S**. This function works as follows:

1. The gNodeB RLC layer evaluates the average processing delay of uplink packets carried by L4S bearers and periodically reports the evaluation results to the gNodeB PDCP layer at intervals specified by the **gNBQciBearer.L4sCongestMarkPeriod** parameter.

2. The **gNBQciBearer.L4sULCongestCheckCoeff** parameter can be adjusted to control the degree of the impact of the historical average delay on the current average delay of uplink packets carried by L4S bearers.
 - If **gNBQciBearer.L4sULCongestCheckCoeff** is set to **1**, the current average delay is not affected by the historical average delay of uplink packets.
 - If **gNBQciBearer.L4sULCongestCheckCoeff** is set to a value greater than **1**, a larger value indicates a greater impact of the historical average delay of uplink packets on the current average delay.
3. The gNodeB PDCP layer compares the evaluated uplink delay with the L4S uplink delay lower threshold (specified by **gNBQciBearer.L4sULDelayLowerThld**) and L4S uplink delay upper threshold (specified by **gNBQciBearer.L4sULDelayUpperThld**). The comparison result is then used by the gNodeB to determine whether congestion has occurred and whether to set the explicit congestion notification (ECN) field in the uplink IP packet headers to **11**.
 - If the evaluated uplink delay is greater than the L4S uplink delay upper threshold and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of the current and all subsequent uplink packets carried by this bearer will be set to **11**.
 - If the evaluated uplink delay is less than or equal to the L4S uplink delay lower threshold, the ECN field in the IP header of the current and all subsequent uplink packets carried by this bearer will not be processed.
 - If the evaluated uplink delay is greater than the L4S uplink delay lower threshold but less than or equal to the L4S uplink delay upper threshold and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of some of the current and subsequent uplink packets carried by this bearer will be set to **11** based on the uplink congestion ratio.
4. After receiving the packets with the ECN field set to **11**, the server instructs the UE to reduce the service rate in order to reduce uplink delay and relieve uplink congestion.

NOTE

The L4S uplink delay lower threshold must be less than or equal to the L4S uplink delay upper threshold.

Uplink congestion ratio = (Evaluated uplink delay – L4S uplink delay lower threshold)/(L4S uplink delay upper threshold – L4S uplink delay lower threshold)

The reporting interval of an uplink L4S UE is specified by the **gNBQciBearer.L4sCongestMarkPeriod** parameter of the bearer with the highest priority for the UE.

Downlink L4S Congestion Control

If downlink congestion control is required, enable the downlink L4S congestion control function, which is controlled by the **DL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter. This function also requires that the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter be set to **L4S**. This function works as follows:

1. The gNodeB RLC layer evaluates the average processing delay of downlink packets carried by L4S bearers and periodically reports the evaluation results

to the gNodeB PDCP layer at intervals specified by the **gNBQciBearer.L4sCongestMarkPeriod** parameter.

2. The **gNBQciBearer.L4sDlCongestCheckCoeff** parameter can be adjusted to control the degree of the impact of the historical average delay on the current average delay of downlink packets carried by L4S bearers.
 - If **gNBQciBearer.L4sDlCongestCheckCoeff** is set to **1**, the current average delay is not affected by the historical average delay of downlink packets.
 - If **gNBQciBearer.L4sDlCongestCheckCoeff** is set to a value greater than **1**, a larger value indicates a greater impact of the historical average delay of downlink packets on the current average delay.
3. The gNodeB PDCP layer compares the evaluated downlink delay with the L4S downlink delay lower threshold (specified by **gNBQciBearer.L4sDlDelayLowerThld**) and L4S downlink delay upper threshold (specified by **gNBQciBearer.L4sDlDelayUpperThld**). The comparison result is then used by the gNodeB to determine whether congestion has occurred and whether to set the ECN field in the downlink IP packet headers to **11**.
 - If the evaluated downlink delay is greater than the L4S downlink delay upper threshold and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of the current and all subsequent downlink packets carried by this bearer will be set to **11**.
 - If the evaluated downlink delay is less than or equal to the L4S downlink delay lower threshold, the ECN field in the IP header of the current and all subsequent downlink packets carried by this bearer will not be processed.
 - If the evaluated downlink delay is greater than the L4S downlink delay lower threshold but less than or equal to the L4S downlink delay upper threshold and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of some of the current and subsequent downlink packets carried by this bearer will be set to **11** based on the downlink congestion ratio.
4. After receiving the packets with the ECN field set to **11**, the UE instructs the server to reduce the service rate in order to reduce downlink delay and relieve downlink congestion.

NOTE

The L4S downlink delay lower threshold must be less than or equal to the L4S downlink delay upper threshold.

Downlink congestion ratio = (Evaluated downlink delay – L4S downlink delay lower threshold)/(L4S downlink delay upper threshold – L4S downlink delay lower threshold)

The reporting interval of a downlink L4S UE is specified by the **gNBQciBearer.L4sCongestMarkPeriod** parameter of the bearer with the highest priority for the UE.

4.1.2 Network Analysis

4.1.2.1 Benefits

L4S congestion control based on local identification reduces the uplink and downlink latency of UEs, improving user experience.

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

After L4S congestion control based on local identification is enabled, the **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** decrease.

4.1.3 Requirements

4.1.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
Low-frequency TDD	FOFD-071203	Adaptive Managed Latency	NR0S00AML000	Macro gNodeB LampSite gNodeB	per gNodeB

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
L4S	gNBQciBearer.L4sFlag set to TRUE	<i>Adaptive Managed Latency</i>	L4S congestion control based on local identification depends on the L4S function.

4.1.3.2.2 Mutually Exclusive Functions

None

4.1.3.2.3 Function Impacts

None

4.1.3.3 Hardware

The hardware described in this section may require compatibility between each other. For details, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. The LBBPd board on the LTE side does not support this function. For details, see technical specifications of the related BBUs in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All RF modules support this function.

Cells

None

4.1.3.4 Others

None

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

4.1.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

For details about activation configurations in NSA Option 3x networking, see *EPC-based NSA Basics*.

Table 4-1 Parameters used for activation (in SA networking)

Parameter Name	Parameter ID	Option	Setting Notes
L4S Flag	<i>gNBQciBearer.L4sFlag</i>	None	To enable the L4S function, set this parameter to TRUE .
L4S Algorithm Switch	<i>gNodeBParam.L4sAlgoSwitch</i>	UL_CONGEST_CTRL_SW	To enable uplink L4S congestion control, select this option.
L4S Algorithm Switch	<i>gNodeBParam.L4sAlgoSwitch</i>	DL_CONGEST_CTRL_SW	To enable downlink L4S congestion control, select this option.
L4S Congestion Detection Algorithm	<i>gNBQciBearer.L4sCongestDetectAlgorithm</i>	None	This parameter specifies the congestion detection algorithm for L4S-related functions. Use the recommended value.
L4S Congestion Mark Period	<i>gNBQciBearer.L4sCongestMarkPeriod</i>	None	This parameter specifies the L4S congestion mark period. Use the recommended value.
L4S Uplink Congestion Check Coefficient	<i>gNBQciBearer.L4sULCongestCheckCoefficient</i>	None	This parameter specifies the coefficient for uplink L4S congestion check. Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
L4S Downlink Congestion Check Coefficient	gNBQciBearer.L4sDlCongestCheckCoefficient	None	This parameter specifies the coefficient for downlink L4S congestion check. Use the recommended value.
L4S Uplink Delay Lower Threshold	gNBQciBearer.L4sUlDelayLowerThreshold	None	Use the recommended value.
L4S Uplink Delay Upper Threshold	gNBQciBearer.L4sUlDelayUpperThreshold	None	Use the recommended value.
L4S Downlink Delay Lower Threshold	gNBQciBearer.L4sDlDelayLowerThreshold	None	Use the recommended value.
L4S Downlink Delay Upper Threshold	gNBQciBearer.L4sDlDelayUpperThreshold	None	Use the recommended value.

4.1.4.1.2 Using MML Commands

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The following configurations apply only to SA networking. For activation command examples in NSA Option 3x networking, see *EPC-based NSA Basics*.

- Activation command examples for uplink L4S congestion control:

```
//Turning on the uplink congestion control switch if uplink congestion control is required
MOD GNODEBPARAM:L4sAlgoSwitch=UL_CONGEST_CTRL_SW-1;

//Configuring L4S Flag
MOD GNBQCIBEARER:Qci=100,L4sFlag=TRUE;

//Configuring the L4S congestion detection algorithm
MOD GNBQCIBEARER:Qci=100,L4sCongestDetectAlgorithm=L4S;

//Configuring the L4S congestion mark period
```

```
MOD GNBQCIBEARER:Qci=100,L4sCongestMarkPeriod=10;  
  
//Configuring the coefficient for uplink L4S congestion check  
MOD GNBQCIBEARER:Qci=100,L4sULCongestCheckCoeff=1;  
  
//Configuring L4S Uplink Delay Lower Threshold  
MOD GNBQCIBEARER:Qci=100,L4sULDelayLowerThld=20;  
  
//Configuring L4S Uplink Delay Upper Threshold  
MOD GNBQCIBEARER:Qci=100,L4sULDelayUpperThld=60;
```

- Activation command examples for downlink L4S congestion control:

```
//Turning on the downlink congestion control switch if downlink congestion control is required  
MOD GNODEBPARAM:L4sAlgoSwitch=DL_CONGEST_CTRL_SW-1;  
  
//Configuring L4S Flag  
MOD GNBQCIBEARER:Qci=100,L4sFlag=TRUE;  
  
//Configuring the L4S congestion detection algorithm  
MOD GNBQCIBEARER:Qci=100,L4sCongestDetectAlgorithm=L4S;  
  
//Configuring the L4S congestion mark period  
MOD GNBQCIBEARER:Qci=100,L4sCongestMarkPeriod=10;  
  
//Configuring the coefficient for downlink L4S congestion check  
MOD GNBQCIBEARER:Qci=100,L4sDLCongestCheckCoeff=1;  
  
//Configuring L4S Downlink Delay Lower Threshold  
MOD GNBQCIBEARER:Qci=100,L4sDLDelayLowerThld=20;  
  
//Configuring L4S Downlink Delay Upper Threshold  
MOD GNBQCIBEARER:Qci=100,L4sDLDelayUpperThld=60;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Deactivation command examples for uplink L4S congestion control:

```
//Setting L4S Flag to FALSE  
MOD GNBQCIBEARER:Qci=100,L4sFlag=FALSE;  
//Turning off the uplink congestion control switch  
MOD GNODEBPARAM:L4sAlgoSwitch=UL_CONGEST_CTRL_SW-0;
```

- Deactivation command examples for downlink L4S congestion control:

```
//Setting L4S Flag to FALSE  
MOD GNBQCIBEARER:Qci=100,L4sFlag=FALSE;  
//Turning off the downlink congestion control switch  
MOD GNODEBPARAM:L4sAlgoSwitch=DL_CONGEST_CTRL_SW-0;
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

Use a packet capture tool to capture uplink and downlink IP packets at the egress of the base station. If the ECN field in an IP packet header is set to **11**, the base

station has set the L4S congestion flag. This means that the L4S congestion control function has taken effect. [Table 4-2](#) describes the ECN value information.

Table 4-2 ECN value information

Binary Codepoint	Codepoint Name	Meaning
00	Non-ECT	Not ECN-capable Transport
01	ECT(1)	L4S-Capable Transport
10	ECT(0)	ECN-Capable Transport
11	CE	Congestion Experienced

4.1.4.3 Network Monitoring

After L4S congestion control is enabled, you can monitor the activation status of this function based on the performance counters listed in [Table 4-3](#).

Table 4-3 Performance counters (in SA networking)

Counter ID	Counter Name
1911833758	N.PDCP.DL.Cell5QI.Delay
1911833755	N.RLC.DL.Cell5QI.Delay
1911833757	N.MAC.DL.Cell5QI.Delay
1911833760	N.PDCP.UL.Cell5QI.Delay
1911833762	N.RLC.UL.Cell5QI.Delay
1911833761	N.MAC.UL.Cell5QI.Delay
1911847107	N.PDCP.UL.L4S.ECT1.Packets.Cell5QI
1911847108	N.PDCP.UL.L4S.CE.Packets.Cell5QI
1911847109	N.PDCP.DL.L4S.ECT1.Packets.Cell5QI
1911847110	N.PDCP.DL.L4S.CE.Packets.Cell5QI

4.2 Enhanced L4S Congestion Control

4.2.1 Principles

The enhanced L4S congestion control function is used to relieve congestion and further reduce the delay of delay-sensitive services such as cloud gaming and XR, thereby improving user experience. The overall process of enhanced L4S

congestion control is the same as that of the L4S congestion control function. The following focuses on the difference between these two functions, which lies in the delay evaluation method.

Enhanced L4S congestion control includes the **Enhanced Uplink L4S Congestion Control** and **Enhanced Downlink L4S Congestion Control** functions. Before enabling enhanced uplink L4S congestion control or enhanced downlink L4S congestion control, enable the L4S function by setting the **gNBQciBearer.L4sFlag** parameter to **TRUE**, and set the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter to **L4SPLUS**.

Enhanced Uplink L4S Congestion Control

If uplink congestion control is required, enable enhanced uplink L4S congestion control, which is controlled by the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter and the **UL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter.

This function works as follows:

1. The gNodeB RLC layer evaluates the average processing delay of uplink packets carried by L4S bearers based on the target rate and periodically reports the evaluation results to the gNodeB PDCP layer at intervals specified by the **gNBQciBearer.L4sCongestMarkPeriod** parameter.
 - The following table describes the parameters that can be used to effectively evaluate the uplink delay.

Parameter Name	Parameter ID	Description
L4S Plus Uplink Relative Priority Weight	gNBQciBearer.L4sPlusULRelativePriWeight	This parameter specifies the weight that is used to be multiplied by the basic scheduling priority of uplink L4S bearers.
L4S Plus Uplink Target Rate Offset	gNBQciBearer.L4sPlusULTargetRateOffset	This parameter specifies the offset to the target rate of uplink L4S bearers. The target rate is the sum of the fair rate and the offset.
L4S Plus Uplink Minimum Rate	gNBQciBearer.L4sPlusULMinRate	This parameter specifies the minimum uplink rate of L4S bearers.
L4S Plus Uplink Headroom Minimum Percentage	gNBQciBearer.L4sPlusULHeadroomMinPct	This parameter ensures a margin between the maximum rate and target rate of uplink L4S bearers. Specifically, the target rate cannot exceed the maximum rate multiplied by the value of this parameter.

Parameter Name	Parameter ID	Description
L4S Plus Maximum Duration	gNBQciBearer.L4sPlusMaxDuration	This parameter specifies the maximum duration in which uplink L4S bearers use the minimum rate.
L4S Plus Minimum Time Interval	gNBQciBearer.L4sPlusMinIntervalTime	This parameter specifies the minimum interval at which uplink L4S bearers use the minimum rate.

NOTE

- Maximum rate: indicates the rate estimated based on the assumption that available RBs are allocated based on the ratio of relative priority weights for L4SPlus.
 - Fair rate: indicates the rate estimated based on the assumption that available RBs are evenly allocated.
 - Target rate: indicates the rate obtained by adding an offset to the fair rate.
2. The **gNBQciBearer.L4sULCongestCheckCoeff** parameter can be adjusted to control the degree of the impact of the historical average delay on the current average delay of uplink packets carried by L4S bearers.
 - If **gNBQciBearer.L4sULCongestCheckCoeff** is set to **1**, the current average delay is not affected by the historical average delay of uplink packets.
 - If **gNBQciBearer.L4sULCongestCheckCoeff** is set to a value greater than **1**, a larger value indicates a greater impact of the historical average delay of uplink packets on the current average delay.
 3. The gNodeB PDCP layer obtains the uplink congestion ratio by comparing the evaluated uplink delay with the L4S uplink delay lower threshold (specified by **gNBQciBearer.L4sUIDelayLowerThld**) and L4S uplink delay upper threshold (specified by **gNBQciBearer.L4sUIDelayUpperThld**).

NOTE

Uplink congestion ratio = (Evaluated uplink delay – L4S uplink delay lower threshold)/(L4S uplink delay upper threshold – L4S uplink delay lower threshold)

Enhanced Downlink L4S Congestion Control

If downlink congestion control is required, enable enhanced downlink L4S congestion control, which is controlled by the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter and the **DL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter.

This function works as follows:

1. The gNodeB RLC layer evaluates the average processing delay of downlink packets carried by L4S bearers based on the target rate and periodically

reports the evaluation results to the gNodeB PDCP layer at intervals specified by the **gNBQciBearer.L4sCongestMarkPeriod** parameter.

- The following table describes the parameters that can be used to effectively evaluate the downlink delay.

Parameter Name	Parameter ID	Description
L4S Plus Downlink Relative Priority Weight	gNBQciBearer.L4sPlus DLRelativePriWeight	This parameter specifies the weight that is used to be multiplied by the basic scheduling priority of downlink L4S bearers.
L4S Plus Downlink Target Rate Offset	gNBQciBearer.L4sPlus DLTargetRateOffset	This parameter specifies the offset to the target rate of downlink L4S bearers. The target rate is the sum of the fair rate and the offset.
L4S Plus Downlink Minimum Rate	gNBQciBearer.L4sPlus DLMinRate	This parameter specifies the minimum downlink rate of L4S bearers.
L4S Plus Downlink Headroom Minimum Percentage	gNBQciBearer.L4sPlus DLHeadroomMinPct	This parameter ensures a margin between the maximum rate and target rate of downlink L4S bearers. Specifically, the target rate cannot exceed the maximum rate multiplied by the value of this parameter.
L4S Plus Maximum Duration	gNBQciBearer.L4sPlus MaxDuration	This parameter specifies the maximum duration in which downlink L4S bearers use the minimum rate.
L4S Plus Minimum Time Interval	gNBQciBearer.L4sPlus MinIntervalTime	This parameter specifies the minimum interval at which downlink L4S bearers use the minimum rate.

NOTE

- Maximum rate: indicates the rate estimated based on the assumption that available RBs are allocated based on the ratio of relative priority weights for L4SPlus.
- Fair rate: indicates the rate estimated based on the assumption that available RBs are evenly allocated.
- Target rate: indicates the rate obtained by adding an offset to the fair rate.

2. The **gNBQciBearer.L4sDlCongestCheckCoeff** parameter can be adjusted to control the degree of the impact of the historical average delay on the current average delay of downlink packets carried by L4S bearers.
 - If **gNBQciBearer.L4sDlCongestCheckCoeff** is set to **1**, the current average delay is not affected by the historical average delay of downlink packets.
 - If **gNBQciBearer.L4sDlCongestCheckCoeff** is set to a value greater than **1**, a larger value indicates a greater impact of the historical average delay of downlink packets on the current average delay.
3. The gNodeB PDCP layer obtains the downlink congestion ratio by comparing the evaluated downlink delay with the L4S downlink delay lower threshold (specified by **gNBQciBearer.L4sDlDelayLowerThld**) and L4S downlink delay upper threshold (specified by **gNBQciBearer.L4sDlDelayUpperThld**).

 NOTE

Downlink congestion ratio = (Evaluated downlink delay – L4S downlink delay lower threshold)/(L4S downlink delay upper threshold – L4S downlink delay lower threshold)

4.2.2 Network Analysis

4.2.2.1 Benefits

In the case of congestion, the enhanced L4S congestion control function further reduces the uplink and downlink latency of UEs and improves user experience on the basis of the L4S congestion control function.

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

In the case of congestion, after the enhanced L4S congestion control function is enabled, the **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** decrease.

4.2.3 Requirements

4.2.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
Low-frequency TDD	FOFD-071203	Adaptive Managed Latency	NR0S00AM L000	Macro gNode B LampSite gNode B	per gNodeB

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
L4S	gNBQciBearer.L4sFlag set to TRUE	<i>Adaptive Managed Latency</i>	Enhanced L4S congestion control depends on the L4S function.

4.2.3.2.2 Mutually Exclusive Functions

None

4.2.3.2.3 Function Impacts

None

4.2.3.3 Hardware

The requirements are the same as those for L4S congestion control based on local identification. For details, see [4.1.3.3 Hardware](#).

4.2.3.4 Others

None

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

Table 4-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-4 Parameters used for activation (in SA networking)

Parameter Name	Parameter ID	Option	Setting Notes
L4S Flag	gNBQciBearer.L4sFlag	None	To enable the L4S function, set this parameter to TRUE .
L4S Algorithm Switch	gNodeBParam.L4sAlgoSwitch	UL_CONGEST_CTRL_SW	To enable uplink L4S congestion control, select this option.
L4S Algorithm Switch	gNodeBParam.L4sAlgoSwitch	DL_CONGEST_CTRL_SW	To enable downlink L4S congestion control, select this option.
L4S Congestion Detection Algorithm	gNBQciBearer.L4sCongestDetectAlgorithm	None	This parameter specifies the congestion detection algorithm for L4S-related functions. The recommended value is L4SPLUS .
L4S Congestion Mark Period	gNBQciBearer.L4sCongestMarkPeriod	None	This parameter specifies the L4S congestion mark period. Use the recommended value.
L4S Uplink Congestion Check Coefficient	gNBQciBearer.L4sUlCongestCheckCoefficient	None	This parameter specifies the coefficient for uplink L4S congestion check. Use the recommended value.
L4S Downlink Congestion Check Coefficient	gNBQciBearer.L4sDlCongestCheckCoefficient	None	This parameter specifies the coefficient for downlink L4S congestion check. Use the recommended value.
L4S Uplink Delay Lower Threshold	gNBQciBearer.L4sUlDelayLowerThreshold	None	Use the recommended value.
L4S Uplink Delay Upper Threshold	gNBQciBearer.L4sUlDelayUpperThreshold	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
L4S Downlink Delay Lower Threshold	<i>gNBQciBearer.L4sDlDelayLowerThld</i>	None	Use the recommended value.
L4S Downlink Delay Upper Threshold	<i>gNBQciBearer.L4sDlDelayUpperThld</i>	None	Use the recommended value.
L4S Plus Uplink Relative Priority Weight	<i>gNBQciBearer.L4sPlusUlRelativePri-Weight</i>	None	Use the recommended value.
L4S Plus Downlink Relative Priority Weight	<i>gNBQciBearer.L4sPlusDlRelativePri-Weight</i>	None	Use the recommended value.
L4S Plus Uplink Target Rate Offset	<i>gNBQciBearer.L4sPlusUlTargetRateOffset</i>	None	Use the recommended value.
L4S Plus Downlink Target Rate Offset	<i>gNBQciBearer.L4sPlusDlTargetRateOffset</i>	None	Use the recommended value.
L4S Plus Uplink Minimum Rate	<i>gNBQciBearer.L4sPlusUlMinRate</i>	None	Use the recommended value.
L4S Plus Downlink Minimum Rate	<i>gNBQciBearer.L4sPlusDlMinRate</i>	None	Use the recommended value.
L4S Plus Uplink Headroom Minimum Percentage	<i>gNBQciBearer.L4sPlusUlHeadroomMinPct</i>	None	Use the recommended value.
L4S Plus Downlink Headroom Minimum Percentage	<i>gNBQciBearer.L4sPlusDlHeadroomMinPct</i>	None	Use the recommended value.
L4S Plus Maximum Duration	<i>gNBQciBearer.L4sPlusMaxDuration</i>	None	Use the recommended value.
L4S Plus Minimum Time Interval	<i>gNBQciBearer.L4sPlusMinInterval-Time</i>	None	Use the recommended value.

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to [4.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Activation command examples for enhanced uplink L4S congestion control:
//Turning on the uplink congestion control switch if uplink congestion control is required
MOD GNODEBPARAM:L4sAlgoSwitch=UL_CONGEST_CTRL_SW-1;

//Configuring the L4S flag
MOD GNBQCIBEARER:Qci=100,L4sFlag=TRUE;

//Configuring the L4S congestion detection algorithm
MOD GNBQCIBEARER:Qci=100,L4sCongestDetectAlgorithm=L4SPLUS;

//Configuring the L4S congestion mark period
MOD GNBQCIBEARER:Qci=100,L4sCongestMarkPeriod=10;

//Configuring the coefficient for uplink L4S congestion check
MOD GNBQCIBEARER:Qci=100,L4sULCongestCheckCoeff=1;

//Configuring the L4S uplink delay lower threshold
MOD GNBQCIBEARER:Qci=100,L4sULDelayLowerThld=20;

//Configuring the L4S uplink delay upper threshold
MOD GNBQCIBEARER:Qci=100,L4sULDelayUpperThld=60;

//Configuring the L4sPlusULRelativePriWeight parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusULRelativePriWeight=2;

//Configuring the L4sPlusULTargetRateOffset parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusULTargetRateOffset=10;

//Configuring the L4sPlusULMinRate parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusULMinRate=1024;

//Configuring the L4sPlusULHeadroomMinPct parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusULHeadroomMinPct=30;

//Configuring the L4sPlusMaxDuration parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusMaxDuration=10;

//Configuring the L4sPlusMinIntervalTime parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusMinIntervalTime=10;
- Activation command examples for enhanced downlink L4S congestion control:
//Turning on the downlink congestion control switch if downlink congestion control is required
MOD GNODEBPARAM:L4sAlgoSwitch=DL_CONGEST_CTRL_SW-1;

//Configuring the L4S flag
MOD GNBQCIBEARER:Qci=100,L4sFlag=TRUE;

//Configuring the L4S congestion detection algorithm
MOD GNBQCIBEARER:Qci=100,L4sCongestDetectAlgorithm=L4SPLUS;

//Configuring the L4S congestion mark period

```
MOD GNBQCIBEARER:Qci=100,L4sCongestMarkPeriod=10;

//Configuring the coefficient for downlink L4S congestion check
MOD GNBQCIBEARER:Qci=100,L4sDlCongestCheckCoeff=1;

//Configuring the L4S downlink delay lower threshold
MOD GNBQCIBEARER:Qci=100,L4sDlDelayLowerThld=20;

//Configuring the L4S downlink delay upper threshold
MOD GNBQCIBEARER:Qci=100,L4sDlDelayUpperThld=60;

//Configuring the L4sPlusDlRelativePriWeight parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusDlRelativePriWeight=2;

//Configuring the L4sPlusDlTargetRateOffset parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusDlTargetRateOffset=10;

//Configuring the L4sPlusDlMinRate parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusDlMinRate=2048;

//Configuring the L4sPlusDlHeadroomMinPct parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusDlHeadroomMinPct=30;

//Configuring the L4sPlusMaxDuration parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusMaxDuration=10;

//Configuring the L4sPlusMinIntervalTime parameter
MOD GNBQCIBEARER:Qci=100,L4sPlusMinIntervalTime=10;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Deactivation command examples for enhanced uplink L4S congestion control:

```
//Setting L4S Flag to FALSE
MOD GNBQCIBEARER:Qci=100,L4sFlag=FALSE;
//Turning off the uplink congestion control switch
MOD GNODEBPARAM:L4sAlgoSwitch=UL_CONGEST_CTRL_SW-0;
```
- Deactivation command examples for enhanced downlink L4S congestion control:

```
//Setting L4S Flag to FALSE
MOD GNBQCIBEARER:Qci=100,L4sFlag=FALSE;
//Turning off the downlink congestion control switch
MOD GNODEBPARAM:L4sAlgoSwitch=DL_CONGEST_CTRL_SW-0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

Use a packet capture tool to capture uplink and downlink IP packets at the egress of the base station. If the ECN field in an IP packet header is set to **11**, the base station has set the L4S congestion flag. This means that the L4S congestion control function has taken effect. [Table 4-5](#) describes the ECN value information.

Table 4-5 ECN value information

Binary Codepoint	Codepoint Name	Meaning
00	Non-ECT	Not ECN-capable Transport
01	ECT(1)	L4S-Capable Transport
10	ECT(0)	ECN-Capable Transport
11	CE	Congestion Experienced

4.2.4.3 Network Monitoring

After enhanced L4S congestion control is enabled, you can monitor its activation status based on the performance counters listed in [Table 4-6](#).

Table 4-6 Performance counters (in SA networking)

Counter ID	Counter Name
1911833758	N.PDCP.DL.Cell5QI.Delay
1911833755	N.RLC.DL.Cell5QI.Delay
1911833757	N.MAC.DL.Cell5QI.Delay
1911833760	N.PDCP.UL.Cell5QI.Delay
1911833762	N.RLC.UL.Cell5QI.Delay
1911833761	N.MAC.UL.Cell5QI.Delay
1911847107	N.PDCP.UL.L4S.ECT1.Packets.Cell5QI
1911847108	N.PDCP.UL.L4S.CE.Packets.Cell5QI
1911847109	N.PDCP.DL.L4S.ECT1.Packets.Cell5QI
1911847110	N.PDCP.DL.L4S.CE.Packets.Cell5QI

4.3 Enhanced L4S Congestion Control Based on Rate Adjustment

4.3.1 Principles

The "enhanced L4S congestion control based on rate adjustment" function adjusts the transmission rates of L4S UEs toward the target values while mitigating congestion, ensuring that the resources for large-packet UEs with service requirements are not preempted by L4S UEs. This function follows the same overall process as the enhanced L4S congestion control function. The difference is

that enhanced L4S congestion control based on rate adjustment uses a virtual queue mechanism to adjust the transmission rate and considers the experience of large-packet UEs in the cell when calculating the maximum rate, fair rate, and target rate to ensure fairness in rate adjustment. The following focuses on this difference.

Enhanced L4S congestion control based on rate adjustment includes the **Enhanced Uplink L4S Congestion Control Based on Rate Adjustment** and **Enhanced Downlink L4S Congestion Control Based on Rate Adjustment** functions. Before enabling these two functions, enable the L4S function by setting the **gNBQciBearer.L4sFlag** parameter to **TRUE**, and set the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter to **L4SPLUS**.

Enhanced Uplink L4S Congestion Control Based on Rate Adjustment

If uplink congestion control based on rate adjustment is required, enable enhanced uplink L4S congestion control (by selecting the **UL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter and setting the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter to **L4SPLUS**), and select the **UL_L4SPLUS_FOR_FAIR_SW** option of the **NRDUCellUISch.UISchAlgoSwitch** parameter.

This function works as follows:

1. The gNodeB RLC layer evaluates the average processing delay of uplink packets carried by L4S bearers based on the target rate and periodically reports the evaluation results to the gNodeB PDCP layer at an interval specified by the **gNBQciBearer.L4sCongestMarkPeriod** parameter.
 - The following table describes the parameters that can be used to effectively evaluate the uplink delay.

Parameter Name	Parameter ID	Description
L4S Plus Uplink Relative Priority Weight	gNBQciBearer.L4sPlusULRelativePriWeight	This parameter specifies the weight that is used to be multiplied by the basic scheduling priority of uplink L4S bearers.
L4S Plus Uplink Target Rate Offset	gNBQciBearer.L4sPlusULTargetRateOffset	This parameter specifies the offset to the target rate of uplink L4S bearers. The target rate is the sum of the fair rate and the offset.
L4S Plus Uplink Minimum Rate	gNBQciBearer.L4sPlusULMinRate	This parameter specifies the minimum uplink rate of L4S bearers.

Parameter Name	Parameter ID	Description
L4S Plus Uplink Headroom Minimum Percentage	gNBQciBearer.L4sPlusUlHeadroomMinPct	This parameter ensures a margin between the maximum rate and target rate of uplink L4S bearers. Specifically, the target rate cannot exceed the maximum rate multiplied by the value of this parameter.
L4S Plus Maximum Duration	gNBQciBearer.L4sPlusMaxDuration	This parameter specifies the maximum duration in which uplink L4S bearers use the minimum rate.
L4S Plus Minimum Time Interval	gNBQciBearer.L4sPlusMinIntervalTime	This parameter specifies the minimum interval at which uplink L4S bearers use the minimum rate.
L4S Plus for Fairness UL Buffer Volume Thld	NRDUCellUISch.L4sPlu sForFairULBufVolThld	This parameter is used to determine whether a UE is an uplink large-packet UE. Specifically, if the filtered BSR-indicated uplink data volume exceeds the threshold for a UE, this UE is considered as an uplink large-packet UE. (BSR is short for buffer status report.)

NOTE

- Maximum rate: Indicates the rate estimated based on the assumption that available RBs are allocated based on the overall weight ratio for L4SPlus. In the calculation, only the overall weight of large-packet UEs in the cell is considered. The higher one between the rate corresponding to the remaining RB resources in the cell and this estimated maximum rate is used as the maximum rate. The overall weight consists of the priority weight and the weight factor for calculating the uplink scheduling priority, which is specified by the **gNBDUMacParamGroup.UlSchPriWeightFactor** parameter.
- Fair rate: indicates the rate estimated based on the assumption that available RBs are evenly allocated. Only large-packet UEs in the cell are considered during even allocation of available RBs. The higher one between the rate corresponding to the remaining RB resources in the cell and this estimated fair rate is used as the fair rate.
- Target rate: indicates the rate obtained by adding an offset to the fair rate. A margin is ensured between the maximum rate and target rate.
- The available RBs used for calculating the maximum rate and fair rate are limited by the BWP1 RBs of the UEs.

2. The **gNBQciBearer.L4sULCongestCheckCoeff** parameter can be adjusted to control the degree of the impact of the historical average delay on the current average delay of uplink packets carried by L4S bearers.
 - If **gNBQciBearer.L4sULCongestCheckCoeff** is set to **1**, the current average delay is not affected by the historical average delay of uplink packets.
 - If **gNBQciBearer.L4sULCongestCheckCoeff** is set to a value greater than **1**, a larger value indicates a greater impact of the historical average delay of uplink packets on the current average delay.
3. The gNodeB evaluates the uplink delay based on the constructed data transmission queue and target rate. Uplink delay = Accumulated data volume in the data transmission queue/Target rate

 NOTE

Data transmission queue: It is not an actual data transmission queue, but a virtual queue used for evaluation. In this virtual queue, data enters the queue at the incoming service rate (that is, the UE data rate) and leaves the queue at the target rate. The delay estimated based on this queue can be used to adjust the incoming service rate toward the target rate, so that the delay is relatively stable.

4. The gNodeB PDCP layer obtains the uplink congestion ratio by comparing the evaluated uplink delay with the L4S uplink delay lower threshold (specified by **gNBQciBearer.L4sUIDelayLowerThld**) and L4S uplink delay upper threshold (specified by **gNBQciBearer.L4sUIDelayUpperThld**).

 NOTE

Uplink congestion ratio = (Evaluated uplink delay – L4S uplink delay lower threshold)/ (L4S uplink delay upper threshold – L4S uplink delay lower threshold)

Enhanced Downlink L4S Congestion Control Based on Rate Adjustment

If downlink congestion control based on rate adjustment is required, enable enhanced downlink L4S congestion control (by selecting the **DL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter and setting the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter to **L4SPLUS**), and select the **DL_L4SPLUS_FOR_FAIR_SW** option of the **NRDUCellDISch.DISchAlgoEnhSwitch** parameter.

This function works as follows:

1. The gNodeB RLC layer evaluates the average processing delay of downlink packets carried by L4S bearers based on the target rate and periodically reports the evaluation results to the gNodeB PDCP layer at an interval specified by the **gNBQciBearer.L4sCongestMarkPeriod** parameter.
 - The following table describes the parameters that can be used to effectively evaluate the downlink delay.

Parameter Name	Parameter ID	Description
L4S Plus Downlink Relative Priority Weight	gNBQciBearer.L4sPlusDLRelativePriWeight	This parameter specifies the weight that is used to be multiplied by the basic scheduling priority of downlink L4S bearers.
L4S Plus Downlink Target Rate Offset	gNBQciBearer.L4sPlusDLTargetRateOffset	This parameter specifies the offset to the target rate of downlink L4S bearers. The target rate is the sum of the fair rate and the offset.
L4S Plus Downlink Minimum Rate	gNBQciBearer.L4sPlusDLMinRate	This parameter specifies the minimum downlink rate of L4S bearers.
L4S Plus Downlink Headroom Minimum Percentage	gNBQciBearer.L4sPlusDLHeadroomMinPct	This parameter ensures a margin between the maximum rate and target rate of downlink L4S bearers. Specifically, the target rate cannot exceed the maximum rate multiplied by the value of this parameter.
L4S Plus Maximum Duration	gNBQciBearer.L4sPlusMaxDuration	This parameter specifies the maximum duration in which downlink L4S bearers use the minimum rate.
L4S Plus Minimum Time Interval	gNBQciBearer.L4sPlusMinIntervalTime	This parameter specifies the minimum interval at which downlink L4S bearers use the minimum rate.
L4S Plus for Fairness DL Buffer Volume Thld	NRDUCellDISch.L4SPlu sForFairDLBufVolThld	This parameter is used to determine whether a UE is a downlink large-packet UE. Specifically, if the amount of filtered downlink data buffered at the RLC layer exceeds the threshold for a UE in a cell, this UE is considered as a downlink large-packet UE.

 NOTE

- Maximum rate: Indicates the rate estimated based on the assumption that available RBs are allocated based on the overall weight ratio for L4SPlus. In the calculation, only the overall weight of large-packet UEs in the cell is considered. The higher one between the rate corresponding to the remaining RB resources in the cell and this estimated maximum rate is used as the maximum rate. The overall weight consists of the priority weight and the weight factor for calculating the downlink scheduling priority. This weight factor is specified by the **gNBDMacParamGroup.DLSchPriWeightFactor** parameter.
 - Fair rate: indicates the rate estimated based on the assumption that available RBs are evenly allocated. Only large-packet UEs in the cell are considered during even allocation of available RBs. The higher one between the rate corresponding to the remaining RB resources in the cell and this estimated fair rate is used as the fair rate.
 - Target rate: indicates the rate obtained by adding an offset to the fair rate. A margin is ensured between the maximum rate and target rate.
 - The available RBs used for calculating the maximum rate and fair rate are limited by the BWP1 RBs of the UEs.
2. The **gNBQciBearer.L4sDlCongestCheckCoeff** parameter can be adjusted to control the degree of the impact of the historical average delay on the current average delay of downlink packets carried by L4S bearers.
- If **gNBQciBearer.L4sDlCongestCheckCoeff** is set to **1**, the current average delay is not affected by the historical average delay of downlink packets.
 - If **gNBQciBearer.L4sDlCongestCheckCoeff** is set to a value greater than **1**, a larger value indicates a greater impact of the historical average delay of downlink packets on the current average delay.
3. The gNodeB evaluates the downlink delay based on the constructed data transmission queue and target rate. Downlink delay = Accumulated data volume in the data transmission queue/Target rate

 NOTE

Data transmission queue: It is not an actual data transmission queue, but a virtual queue used for evaluation. In this virtual queue, data enters the queue at the incoming service rate (that is, the UE data rate) and leaves the queue at the target rate. The delay estimated based on this queue can be used to adjust the incoming service rate toward the target rate, so that the delay is relatively stable.

4. The gNodeB PDCP layer obtains the downlink congestion ratio by comparing the evaluated downlink delay with the L4S downlink delay lower threshold (specified by **gNBQciBearer.L4sDlDelayLowerThld**) and L4S downlink delay upper threshold (specified by **gNBQciBearer.L4sDlDelayUpperThld**).

 NOTE

Downlink congestion ratio = (Evaluated downlink delay – L4S downlink delay lower threshold)/(L4S downlink delay upper threshold – L4S downlink delay lower threshold)

4.3.2 Network Analysis

4.3.2.1 Benefits

In the case of congestion, compared with the enhanced L4S congestion control function, enhanced L4S congestion control based on rate adjustment reduces the

uplink and downlink latency of UEs, improves user experience, and prevents resource preemption by L4S UEs for large-packet UEs.

4.3.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

Under congestion conditions, after enhanced L4S congestion control based on rate adjustment is enabled, the **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** fluctuate.

4.3.3 Requirements

4.3.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
Low-frequency TDD	FOFD-071203	Adaptive Managed Latency	NR0S00AML000	Macro gNodeB LampSite gNodeB	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	L4S	gNBQciBearer.L4sFlag set to TRUE	<i>Adaptive Managed Latency</i>	Enhanced L4S congestion control depends on the L4S function.

4.3.3.2 Mutually Exclusive Functions

None

4.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgorithmSwitch.NetworkSliceAlgorithmSwitch parameter	<i>Network Slicing</i>	If both "NR DU cell resource management between network slices" and "enhanced L4S congestion control based on rate adjustment" are enabled, the gNodeB restricts the available RBs for L4S UEs based on the available resource range of the network slice group to which the UEs belong, which may cause the calculated maximum rate and fair rate to be inaccurate.
Low-frequency TDD	RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode set to SHARED_FREQ	<i>Multi-Operator Sharing</i>	If enhanced L4S congestion control based on rate adjustment is enabled in RAN sharing with common carrier scenarios, the gNodeB restricts the available RBs for L4S UEs based on the available resource range of the UEs' operator, which may cause the calculated maximum rate and fair rate to be inaccurate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	If both "user PRB control for FWA" and "enhanced L4S congestion control based on rate adjustment" are enabled, the gNodeB restricts the available RBs for L4S UEs based on the available resource range of FWA UEs, which may cause the calculated maximum rate and fair rate to be inaccurate.
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	If both "QCI-based resource control" and "enhanced L4S congestion control based on rate adjustment" are enabled, the gNodeB restricts the available RBs for L4S UEs based on the available resources of the QCI resource group associated with the UEs, which may cause the calculated maximum rate and fair rate to be inaccurate.
Low-frequency TDD	MU-MIMO	DL_MU_MIMO_SW and UL_MU_MIMO_SW options of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	If both MU-MIMO and "enhanced L4S congestion control based on rate adjustment" are enabled, the maximum rate and fair rate calculated by the gNodeB for L4S UEs may be inaccurate.

4.3.3.3 Hardware

The requirements are the same as those for L4S congestion control based on local identification. For details, see [4.1.3.3 Hardware](#).

4.3.3.4 Others

None

4.3.4 Operation and Maintenance

4.3.4.1 Data Configuration

4.3.4.1.1 Data Preparation

[Table 4-7](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-7 Parameters used for activation (in SA networking)

Parameter Name	Parameter ID	Option	Setting Notes
L4S Flag	gNBQciBearer.L4sFlag	None	To enable the L4S function, set this parameter to TRUE .
L4S Algorithm Switch	gNodeBParam.L4sAlgoSwitch	UL_CONGEST_CTRL_SW	To enable uplink L4S congestion control, select this option.
L4S Algorithm Switch	gNodeBParam.L4sAlgoSwitch	DL_CONGEST_CTRL_SW	To enable downlink L4S congestion control, select this option.
L4S Congestion Detection Algorithm	gNBQciBearer.L4sCongestDetectAlgorithm	None	This parameter specifies the congestion detection algorithm for L4S-related functions. The recommended value is L4SPLUS .
UL Scheduling Algorithm Switch	NRDUCellUISch.UISchAlgoSwitch	UL_L4SPLUS_FOR_FAIR_SW	To enable enhanced uplink L4S congestion control based on rate adjustment, select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Schedule Algorithm Enhanced Switch	NRDUCellDLSch.DLSchAlgoEnhSwitch	DL_L4SPLUS_FOR_FAIR_SW	To enable enhanced downlink L4S congestion control based on rate adjustment, select this option.
L4S Congestion Mark Period	gNBQciBearer.L4sCongestMarkPeriod	None	This parameter specifies the L4S congestion mark period. Use the recommended value.
L4S Uplink Congestion Check Coefficient	gNBQciBearer.L4sULCongestCheckCoeff	None	This parameter specifies the coefficient for uplink L4S congestion check. Use the recommended value.
L4S Downlink Congestion Check Coefficient	gNBQciBearer.L4sDLCongestCheckCoeff	None	This parameter specifies the coefficient for downlink L4S congestion check. Use the recommended value.
L4S Uplink Delay Lower Threshold	gNBQciBearer.L4sULDelayLowerThreshold	None	Use the recommended value.
L4S Uplink Delay Upper Threshold	gNBQciBearer.L4sULDelayUpperThreshold	None	Use the recommended value.
L4S Downlink Delay Lower Threshold	gNBQciBearer.L4sDLDelayLowerThreshold	None	Use the recommended value.
L4S Downlink Delay Upper Threshold	gNBQciBearer.L4sDLDelayUpperThreshold	None	Use the recommended value.
L4S Plus Uplink Relative Priority Weight	gNBQciBearer.L4sPlusULRelativePriorityWeight	None	Use the recommended value.
L4S Plus Downlink Relative Priority Weight	gNBQciBearer.L4sPlusDLRelativePriorityWeight	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
L4S Plus Uplink Target Rate Offset	gNBQciBearer.L4sPlusULTargetRateOffset	None	Use the recommended value.
L4S Plus Downlink Target Rate Offset	gNBQciBearer.L4sPlusDLTargetRateOffset	None	Use the recommended value.
L4S Plus Uplink Minimum Rate	gNBQciBearer.L4sPlusULMinRate	None	Use the recommended value.
L4S Plus Downlink Minimum Rate	gNBQciBearer.L4sPlusDLMinRate	None	Use the recommended value.
L4S Plus Uplink Headroom Minimum Percentage	gNBQciBearer.L4sPlusULHeadroomMinPct	None	Use the recommended value.
L4S Plus Downlink Headroom Minimum Percentage	gNBQciBearer.L4sPlusDLHeadroomMinPct	None	Use the recommended value.
L4S Plus Maximum Duration	gNBQciBearer.L4sPlusMaxDuration	None	Use the recommended value.
L4S Plus Minimum Time Interval	gNBQciBearer.L4sPlusMinInterval-Time	None	Use the recommended value.
L4S Plus for Fairness UL Buffer Volume Thld	NRDUCellULSch.L4SPlusForFairULBufVolThld	None	Use the recommended value.
L4S Plus for Fairness DL Buffer Volume Thld	NRDUCellDISch.L4SPlusForFairDLBufVolThld	None	Use the recommended value.

4.3.4.1.2 Using MML Commands

Before using MML commands, refer to [4.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Activation command examples for enhanced uplink L4S congestion control:
//Turning on the uplink congestion control switch if uplink congestion control is required
MOD GNODEBPARAM: L4sAlgoSwitch=UL_CONGEST_CTRL_SW-1;

//Configuring the L4S flag
MOD GNBQCIBEARER: Qci=100,L4sFlag=TRUE;

//Configuring the L4S congestion detection algorithm
MOD GNBQCIBEARER: Qci=100,L4sCongestDetectAlgorithm=L4SPLUS;

//Enabling enhanced uplink L4S congestion control based on rate adjustment if fairness-based uplink congestion control is required
MOD NRDUCELLULSCH: NrDuCellId=0,UISchAlgoSwitch=UL_L4SPLUS_FOR_FAIR_SW-1;

//Configuring the L4S congestion mark period
MOD GNBQCIBEARER: Qci=100,L4sCongestMarkPeriod=10;

//Configuring the coefficient for uplink L4S congestion check
MOD GNBQCIBEARER: Qci=100,L4sUICongestCheckCoeff=1;

//Configuring the L4S uplink delay lower threshold
MOD GNBQCIBEARER: Qci=100,L4sUIDelayLowerThld=20;

//Configuring the L4S uplink delay upper threshold
MOD GNBQCIBEARER: Qci=100,L4sUIDelayUpperThld=60;

//Configuring the L4sPlusUIRelativePriWeight parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusUIRelativePriWeight=2;

//Configuring the L4sPlusUITargetRateOffset parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusUITargetRateOffset=10;

//Configuring the L4sPlusUIMinRate parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusUIMinRate=1024;

//Configuring the L4sPlusUIHeadroomMinPct parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusUIHeadroomMinPct=30;

//Configuring the L4sPlusMaxDuration parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusMaxDuration=10;

//Configuring the L4sPlusMinIntervalTime parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusMinIntervalTime=10;

//Configuring the L4SPlusForFairUIBufVolThld parameter
MOD NRDUCELLULSCH: NrDuCellId=0,L4SPlusForFairUIBufVolThld=40;
- Activation command examples for enhanced downlink L4S congestion control:
//Turning on the downlink congestion control switch if downlink congestion control is required
MOD GNODEBPARAM: L4sAlgoSwitch=DL_CONGEST_CTRL_SW-1;

//Configuring the L4S flag
MOD GNBQCIBEARER: Qci=100,L4sFlag=TRUE;

//Configuring the L4S congestion detection algorithm
MOD GNBQCIBEARER: Qci=100,L4sCongestDetectAlgorithm=L4SPLUS;

//Enabling enhanced downlink L4S congestion control based on rate adjustment if fairness-based downlink congestion control is required
MOD NRDUCELLDLSCH: NrDuCellId=0,DISchAlgoEnhSwitch=DL_L4SPLUS_FOR_FAIR_SW-1;

```
//Configuring the L4S congestion mark period
MOD GNBQCIBEARER: Qci=100,L4sCongestMarkPeriod=10;

//Configuring the coefficient for downlink L4S congestion check
MOD GNBQCIBEARER: Qci=100,L4sDlCongestCheckCoeff=1;

//Configuring the L4S downlink delay lower threshold
MOD GNBQCIBEARER: Qci=100,L4sDlDelayLowerThld=20;

//Configuring the L4S downlink delay upper threshold
MOD GNBQCIBEARER: Qci=100,L4sDlDelayUpperThld=60;

//Configuring the L4sPlusDlRelativePriWeight parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusDlRelativePriWeight=2;

//Configuring the L4sPlusDlTargetRateOffset parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusDlTargetRateOffset=10;

//Configuring the L4sPlusDlMinRate parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusDlMinRate=2048;

//Configure the L4sPlusDlHeadroomMinPct parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusDlHeadroomMinPct=30;

//Configuring the L4sPlusMaxDuration parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusMaxDuration=10;

//Configuring the L4sPlusMinIntervalTime parameter
MOD GNBQCIBEARER: Qci=100,L4sPlusMinIntervalTime=10;

//Configuring the L4SPlusForFairDlBufVolThld parameter
MOD NRDUCELLDLSCH: NrDuCellId=0,L4SPlusForFairDlBufVolThld=80;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Deactivation command examples for enhanced uplink L4S congestion control:

```
//Setting L4S Flag to FALSE
MOD GNBQCIBEARER: Qci=100,L4sFlag=FALSE;
//Disabling enhanced uplink L4S congestion control based on rate adjustment
MOD NRDUCELLULSCH: NrDuCellId=0,UlSchAlgoSwitch=UL_L4SPLUS_FOR_FAIR_SW-0;
//Turning off the uplink congestion control switch
MOD GNODEBPARAM: L4sAlgoSwitch=UL_CONGEST_CTRL_SW-0;
```
- Deactivation command examples for enhanced downlink L4S congestion control:

```
//Setting L4S Flag to FALSE
MOD GNBQCIBEARER: Qci=100,L4sFlag=FALSE;
//Disabling enhanced downlink L4S congestion control based on rate adjustment
MOD NRDUCELLDLSCH: NrDuCellId=0,DlSchAlgoEnhSwitch=DL_L4SPLUS_FOR_FAIR_SW-0;
//Turning off the downlink congestion control switch
MOD GNODEBPARAM: L4sAlgoSwitch=DL_CONGEST_CTRL_SW-0;
```

4.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.4.2 Activation Verification

Use a packet capture tool to capture uplink and downlink IP packets at the egress of the base station. If the ECN field in an IP packet header is set to **11**, the base station has set the L4S congestion flag. This means that enhanced L4S congestion control based on rate adjustment has taken effect. [Table 4-8](#) describes the ECN value information.

Table 4-8 ECN value information

Binary Codepoint	Codepoint Name	Meaning
00	Non-ECT	Not ECN-capable Transport
01	ECT(1)	L4S-Capable Transport
10	ECT(0)	ECN-Capable Transport
11	CE	Congestion Experienced

4.3.4.3 Network Monitoring

After enhanced L4S congestion control based on rate adjustment is enabled, you can monitor its activation status based on the performance counters listed in [Table 4-9](#).

Table 4-9 Performance counters (in SA networking)

Counter ID	Counter Name
1911833758	N.PDCP.DL.Cell5QI.Delay
1911833755	N.RLC.DL.Cell5QI.Delay
1911833757	N.MAC.DL.Cell5QI.Delay
1911833760	N.PDCP.UL.Cell5QI.Delay
1911833762	N.RLC.UL.Cell5QI.Delay
1911833761	N.MAC.UL.Cell5QI.Delay
1911847107	N.PDCP.UL.L4S.ECT1.Packets.Cell5QI
1911847108	N.PDCP.UL.L4S.CE.Packets.Cell5QI
1911847109	N.PDCP.DL.L4S.ECT1.Packets.Cell5QI
1911847110	N.PDCP.DL.L4S.CE.Packets.Cell5QI

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR;Base Station (BS) radio transmission and reception"
- *3900 & 5900 Series Base Station Product Documentation*
- *EPC-based NSA Basics*
- *Network Slicing*
- *Multi-Operator Sharing*
- *FWA*
- *QCI-based Resource Control*
- *MIMO (TDD)*